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PENJELASAN PROGRAM LOGIKA FUZZY KIPAS ANGIN

1. **Import Library**

```
import numpy as np
import skfuzzy as fuzz
from skfuzzy import control as ctrl
import matplotlib.pyplot as plt
```

Penjelasan:

- numpy → membuat rentang nilai (0–40, 0–100)
- skfuzzy → membuat fungsi keanggotaan
- control → mengatur sistem fuzzy (rule, input, output)
- matplotlib → menampilkan grafik

2. **Membuat Variabel Fuzzy**

```
suhu = ctrl.Antecedent(np.arange(0, 41, 1), 'suhu')
kelembapan = ctrl.Antecedent(np.arange(0, 101, 1), 'kelembapan')
kipas = ctrl.Consequent(np.arange(0, 101, 1), 'kipas')
```

Penjelasan:

- Antecedent = variabel input
- Consequent = variabel output
- np.arange() = membuat range nilai
- 'suhu', 'kelembapan', 'kipas' = nama variabel

3. **Membership Function (Fungsi Keanggotaan)**

- **Suhu**

```
suhu['dingin'] = fuzz.trimf(suhu.universe, [0, 0, 20])
suhu['normal'] = fuzz.trimf(suhu.universe, [15, 25, 30])
suhu['panas'] = fuzz.trimf(suhu.universe, [25, 40, 40]
```

- **Kelembapan**

```
kelembapan['kering'] = fuzz.trimf(kelembapan.universe, [0, 0, 40])
kelembapan['normal'] = fuzz.trimf(kelembapan.universe, [30, 50, 70])
kelembapan['lembap'] = fuzz.trimf(kelembapan.universe, [60, 100, 100])
```

- **Kipas**

```
kipas['lambat'] = fuzz.trimf(kipas.universe, [0, 0, 40])
kipas['sedang'] = fuzz.trimf(kipas.universe, [30, 50, 70])
kipas['cepat'] = fuzz.trimf(kipas.universe, [60, 100, 100])
```

Penjelasan:

- trimf = fungsi segitiga
- [a, b, c] = titik awal, tengah, akhir
- Digunakan untuk menentukan derajat keanggotaan

4. **Rule Fuzzy**

```
rule1 = ctrl.Rule(suhu['dingin'] & kelembapan['kering'], kipas['lambat'])
rule2 = ctrl.Rule(suhu['normal'] & kelembapan['normal'], kipas['sedang'])
rule3 = ctrl.Rule(suhu['panas'] & kelembapan['lembap'], kipas['cepat'])
rule4 = ctrl.Rule(suhu['panas'] & kelembapan['kering'], kipas['sedang'])
rule5 = ctrl.Rule(suhu['dingin'] & kelembapan['lembap'], kipas['sedang'])
```

Penjelasan:

- ctrl.Rule() = membuat aturan fuzzy
- & = operator AND
- Format: **IF kondisi THEN output**

5. **Sistem Kontrol**

```
kipas_ctrl = ctrl.ControlSystem([rule1, rule2, rule3, rule4, rule5])
simulasi = ctrl.ControlSystemSimulation(kipas_ctrl)
```

Penjelasan:

- ControlSystem = kumpulan rule
- ControlSystemSimulation = menjalankan sistem fuzzy

6. **Input**

```
simulasi.input['suhu'] = suhu_input
simulasi.input['kelembapan'] = kelembapan_input
```

Penjelasan:

- Memasukkan nilai suhu dan kelembapan ke sistem

7. **Proses Fuzzy**

```
simulasi.compute()
```

Penjelasan:

Proses yang terjadi:

1. Fuzzifikasi (mengubah nilai ke derajat keanggotaan)
2. Evaluasi rule
3. Defuzzifikasi (menghasilkan output angka)

8. **Output**

```
hasil = simulasi.output['kipas']
print("\n=== HASIL ===")
print("Kecepatan Kipas:", hasil)
```

Penjelasan:

- Mengambil hasil akhir berupa angka (crisp value)

9. **Visualisasi**

```
suhu.view()
kelembapan.view()
```

```
kipas.view(sim=simulasi)

plt.show()
```

Penjelasan:

- .view() = menampilkan grafik membership
- plt.show() = agar grafik tidak langsung tertutup

```
input("Tekan Enter untuk keluar...")
```

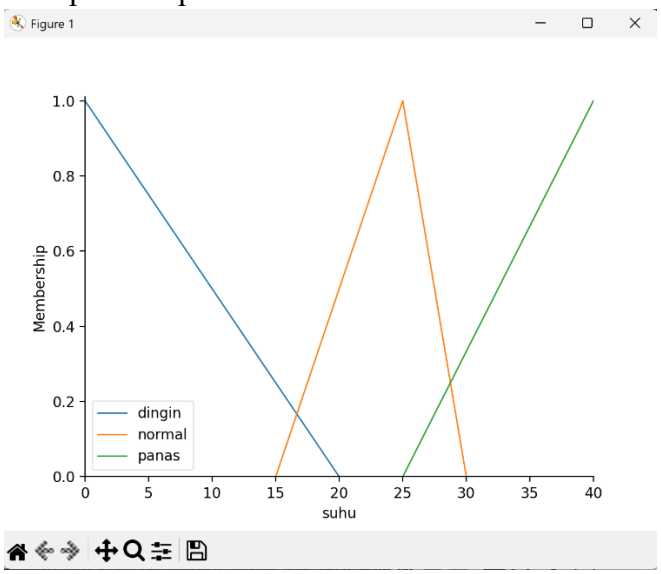
Penjelasan:

- Menahan program supaya tidak langsung tertutup

- **Output Program**

PS C:\Users\ASUS\Documents\semester 4\Praktikum KB\PraktikumKB-Pertemuan2> & C:\Users\ASUS\AppData\Local\Programs\Python\Python312\python.exe "c:/Users/ASUS/Documents/semester 4/Praktikum KB/PraktikumKB-Pertemuan2/Tugas/Kipas Angin.py"
Masukkan suhu (0-40): 30
Masukkan kelembapan (0-100): 80

=== HASIL ===
Kecepatan Kipas: 83.11111111111109



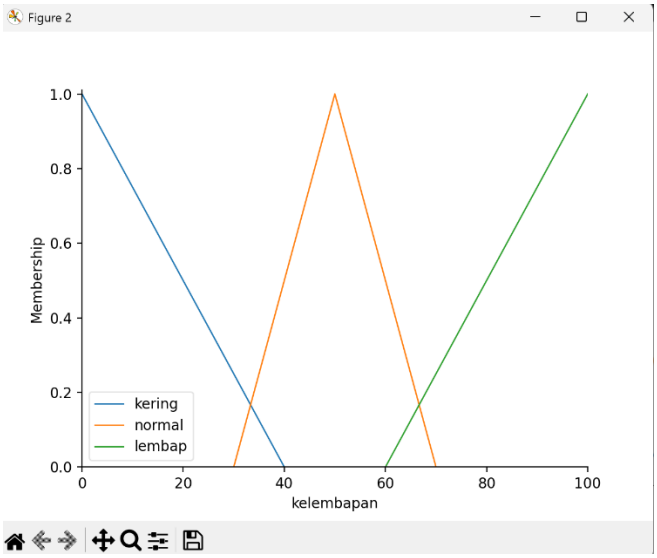


Figure 2
1.0
0.8
0.6
0.4
0.2
0.0
Membership
kering
normal
lembap
0
20
40
60
80
100
kelembapan

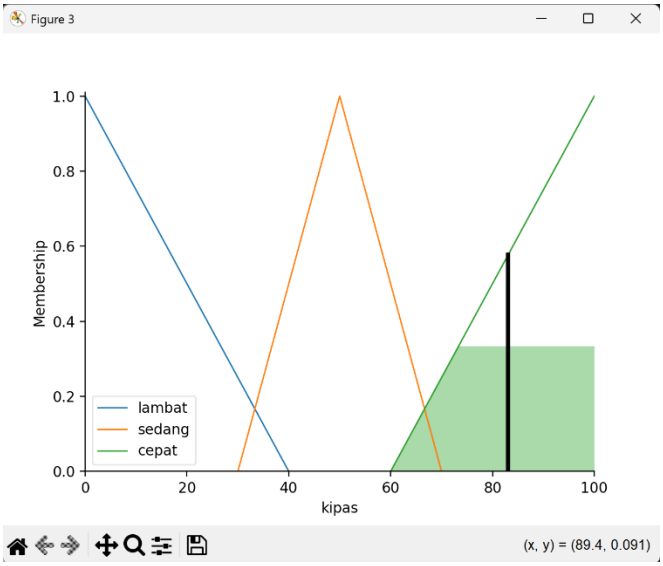


Figure 3
1.0
0.8
0.6
0.4
0.2
0.0
Membership
lambat
sedang
cepat
0
20
40
60
80
100
kipas
(x, y) = (89.4, 0.091)